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Sediment Storage Behind Low-Head Dams: Assessing Potential Contribution to Reservoir Sedimentation in Kansas, USA

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ABSTRACT

Low-head dams (LHDs) are widespread in river systems, yet the volume of sediment impounded behind them remains largely unquantified. Failure or planned removal of these often-aging structures can mobilize stored sediment, posing added risk to reservoirs already losing capacity to sedimentation. In Kansas, the failure of an LHD in May 2018 released sediment equivalent to ~25% of the downstream reservoir's annual accumulation and motivated a broader assessment of storage behind other LHDs. Addressing this knowledge gap, this study quantifies sediment storage behind LHDs located upstream of three federal reservoirs: Tuttle Creek Lake, Perry Lake, and Kanopolis Lake. An integrated approach was employed, combining remote sensing for LHD identification with field-based bathymetric surveys and sediment analysis to directly measure accumulated sediment volumes (V_{LHD}) at representative sites and to estimate V_{LHD} at remotely sensed sites. The relative storage for individual LHDs, expressed as a fraction of downstream annual reservoir sedimentation (f_{ARS}), ranged from 0.005 to 0.659, with a median f_{ARS} of 0.025. Storage volume (V_{LHD}) was more closely related to local physical properties of LHDs—such as dam height and width—than watershed-scale drivers, such as precipitation and drainage area. Bathymetric maps revealed scour holes immediately upstream of LHDs, typically centered 20–50 m upstream and up to ~1.3 m deep, indicating partial sediment continuity and dynamic equilibrium. This research provides crucial data to inform regional sediment management and enhance reservoir sustainability in Kansas and offers a transferable methodology for quantifying the contribution of LHDs in other watersheds.

1 | Introduction

Sedimentation in reservoirs is a major challenge to sustainable water resources management, reducing storage capacity and impairing water quality (Morris and Fan 1998). The accumulation of sediment reduces reservoir storage capacity, degrades water quality, and threatens infrastructure longevity (George et al. 2017; Kondolf et al. 2014). Across the United States and globally, sedimentation has already reduced the effectiveness of many reservoirs, with projected losses accelerating due to aging infrastructure and land use changes (Minear and Kondolf 2009; Randle et al. 2021). In the central U.S., sedimentation is particularly critical for water supply and flood control, as many federal reservoirs are nearing or have surpassed their design life and are experiencing reduced storage capacity (Juracek 2011). Nearly half of the original reservoir capacity in Kansas is expected to be lost over the next 100 years (Rahmani et al. 2018; USACE 2024). Effective sediment management strategies require an understanding of the sources, storage, and transport of sediment throughout a watershed.

Low-head dams (LHDs) are man-made structures that typically span the entire width of a river and are generally less than 5 m in height, allowing water to continuously flow over their crest (Poff and Hotchkiss 2023; Tschantz and Wright 2011). LHDs are widespread across U.S. river systems and often serve as legacy structures from past industrial, agricultural, or municipal uses (Poff and Hart 2002). Despite their modest size, LHDs can accumulate significant sediment volumes over decades (Walter and Merritts 2008). While large reservoirs and dams have been extensively studied for their sediment trapping efficiency and impacts on downstream sediment budgets (Graf 2006; Minear and Kondolf 2009), small impoundments such as LHDs have received far less attention in sediment research (Ralston et al. 2021). As many LHDs are obsolete and increasingly targeted for removal (Csiki and Rhoads 2010; Duda and Bellmore 2022) or at risk of failure (Mansfield et al. 2024), there is a growing need to assess the consequences of sudden sediment release, especially where downstream reservoirs are already sediment impaired (Bellmore et al. 2019).

Although the ecological and hydrological impacts of small dams have gained attention in recent years (Fencl et al. 2015), quantitative assessments of sediment volumes stored behind LHDs have received limited attention. While earlier studies often focused on modeling or large-scale reservoirs (Csiki and Rhoads 2010; Pearson and Pizzuto 2015), recent work has begun to address sediment storage in smaller impoundments and its implications for sediment budgets and downstream sediment delivery (Casserly et al. 2020, 2021; Magilligan et al. 2021; Ramos-Diez et al. 2017; Sindelar et al. 2017). Nonetheless, the number of such studies remains modest relative to the widespread presence of LHDs, and further site-specific, volumetric analyses are needed to support informed sediment management strategies. Understanding sediment storage in LHDs is especially relevant in sediment-sensitive regions like Kansas, where downstream federal reservoirs have already experienced notable storage loss (Rahmani et al. 2018). For example, aerial LiDAR surveys of a recent LHD failure in Marysville, KS, estimated that the volume of sediment released totaled approximately 25% of the annual downstream reservoir sedimentation (Mansfield et al. 2024).

This study quantifies sediment storage behind LHDs in selected Kansas basins to evaluate their potential contribution to downstream reservoir sedimentation (Figure 1). The specific objectives were to: (1) directly measure sediment accumulation volumes and characteristics using field-based bathymetric surveys and sediment sampling at representative LHDs; (2) leverage remote sensing data to estimate sediment storage for LHDs not surveyed in the field; and (3) integrate these direct measurements and estimates to determine the total potential LHD sediment storage within the study basins. Predictive models were developed to estimate the volume of sediment storage behind LHDs (V_{LHD}) using remotely sensed data products. Finally, the significance of this stored sediment was assessed by comparing the potential contribution from LHDs to the annual sediment accumulation rates observed in the downstream federal reservoirs (expressed as a fraction, f_{ARS}).

2 | Study Sites

The study sites are LHDs located within the drainage basins of three federal reservoirs in Kansas (Figure 1): Tuttle Creek Lake, Kanopolis Lake, and Perry Lake. These basins reside in the Kansas River Basin, which experiences a rainfall gradient from 400 mm/year in the west to 1000 mm/year in the east (Lin and Brunsell 2013). The geographical region is classified as lowland plains, with an average slope of 0.065% and a mean annual temperature of approximately 12.78°C, placing it in a temperate climate zone. The land use in these regions is predominantly agricultural, with extensive crop production and pastureland dominating the landscape (Juracek and Mau 2002). This land use pattern has driven the construction of LHDs for water retention, irrigation, and erosion control.

3 | Methods

3.1 | Low-Head Dam Identification

Potential LHD survey sites upstream of the three study reservoirs were identified using a national database compiled by the Low Head Dam Task Force (2024). In the database, initial LHD identification involved visual identification based on satellite imagery along river networks. These preliminary findings were further scrutinized to verify whether the structures were indeed LHDs. Due to the small size of some LHDs and limited satellite imagery resolution, not all LHD-like features could be confirmed as actual LHDs. Thus, two classifications emerge from the Low Head Dam Task Force data set: confirmed LHDs and suspected LHDs (Figure 1).

3.2 | Low-Head Dam Surveying

Field surveys were conducted at seven confirmed LHDs. Sites were prioritized based on public accessibility and property owner permission. Because LHDs pose a potential drowning hazard (McCurry et al. 2023), an unmanned surface vessel (BlueBoat; Blue Robotics Inc., USA) was used to survey the area upstream of each impoundment (Figure 2). The BlueBoat is a compact, lightweight autonomous surface vessel designed for hydrographic and environmental data collection. The vessel was equipped with a sonar altimeter and echosounder, which emits acoustic pulses at 115 kHz for high-resolution depth measurements. The depth measurements were georeferenced using an onboard GPS unit, and a quality control module automatically filtered depth data with a confidence level below 95%. Dam height was measured using a total station, recording the vertical distance from the LHD crest to the downstream streambed. Dam width was determined using satellite imagery (Google Earth). The depth to the sediment surface upstream of the LHD was measured by the BlueBoat; the average depth within the surveyed depositional zone (typically up to 100 m upstream) was used to represent mean depth to sediment.

3.3 | Physical Properties of Sediment Deposits

Sediment cores were collected at each site using a universal coring device (Aquatic Research Instruments, USA). Sediment

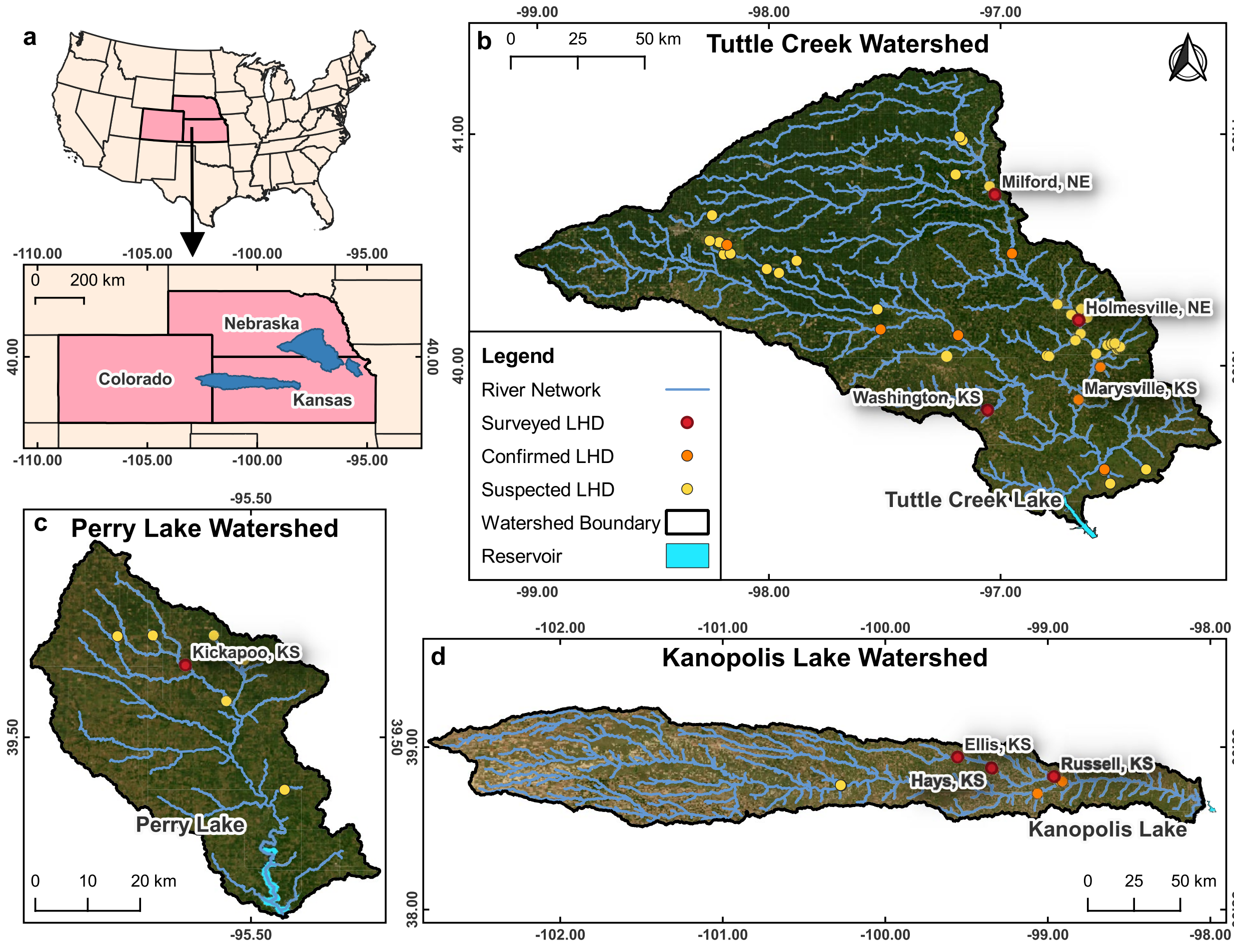

FIGURE 1 | (a) Locations of the three study watersheds that drain into federal reservoirs. The location of surveyed, confirmed, and suspected low-head dams (LHDs) is shown for (b) Tuttle Creek Lake, (c) Perry Lake, and (d) Kanopolis Lake. Surveyed LHDs are labeled based on their proximity to a nearby town or city. Confirmed LHDs are those that are known to exist but were not surveyed. Suspected LHDs appear to be dams from geospatial imagery but were not confirmed with field investigation (Low Head Dam Task Force 2024). The LHD failure at Marysville, KS, is labeled and was surveyed as part of a separate analysis (Mansfield et al. 2024). [Color figure can be viewed at wileyonlinelibrary.com]

from these cores was analyzed for particle size distribution and bulk density. Cores were collected from an accessible and wadable location upstream of the LHD, typically targeting areas presumed to represent recent deposition. Single cores were collected from each LHD, with core depths ranging from 30 to 100 cm. After collection, the samples were transported to the laboratory and stored under refrigeration. For bulk density determination, three evenly spaced subsamples were extracted from each core. Each subsample (2.5 cm interval) was dried at 110°C for 24 h, in accordance with the method used by O'Kelly (2005). The bulk density was determined as the dry weight of sediment divided by the total volume of the subsample. The dry sediment was analyzed for particle size distribution using a Beckman Coulter LS 13320 laser diffraction system (Beckman Coulter Life Sciences, USA).

3.4 | Sediment Volume and Mass Computation

The volume and mass of sediment stored behind each LHD were determined using field survey results, sediment physical properties, and geomorphometric measurements. The volume of sediment deposits trapped behind the LHDs was calculated using the wedge prism formula. The wedge prism formula is commonly applied for estimating storage in small impoundments, assuming a simplified rectangular cross-section and linear upstream thinning of the deposit as:

$$V_{\mathrm{LHD}} = 1/2 \cdot Z \cdot W \cdot L \tag{1}$$

where V_{LHD} is the volume of sediment stored (m^3), Z is the height of sediment deposition upstream from the dam (m), W is the dam width (m), and L is the length of the backwater effect where the presence of the LHD promotes deposition (m).

The backwater length (L), or the distance that the LHD impoundment promoted sediment deposition, was assumed to roughly followed the bed slope, extending the sediment wedge linearly upstream:

$$L = Z/S \tag{2}$$

where S is the natural slope of the riverbed (m/m), which was determined using Google Earth following the approach

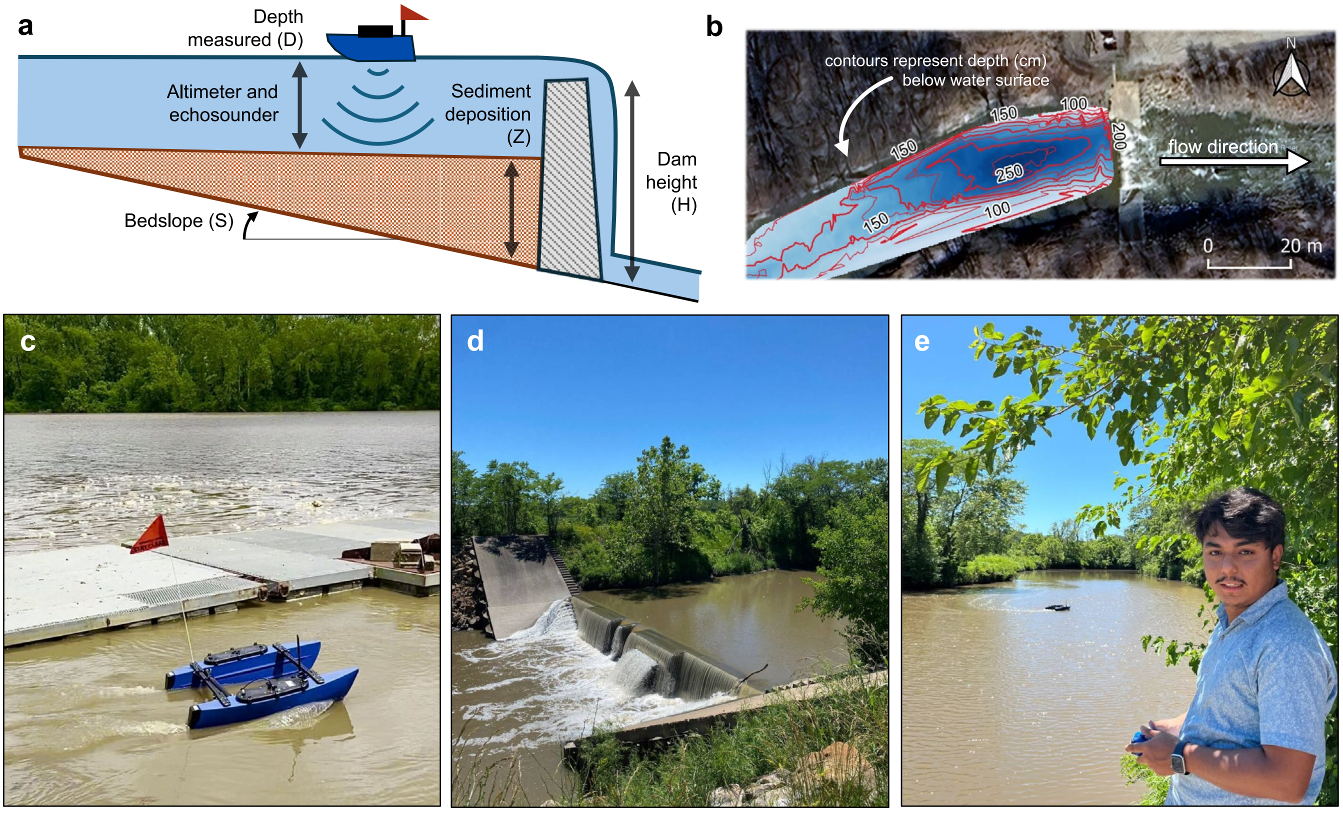

FIGURE 2 | (a) Conceptual diagram of low-head dam (LHD) surveying, demonstrating the dam height, sediment depth, bed slope, and surveying vessel. (b) Bathymetric map showing spatial contours of depth measured (D, in centimeters) upstream of an LHD at Kickapoo, KS. (c) Testing of the surveying vessel on the Kansas River near Lawrence, KS. (d) An example of a run-of-river LHD located at Kickapoo, KS. (e) Deployment of the surveying vessel at Kickapoo, KS. The symbols for dam height (H), sediment depth (Z), and bed slope (S) represent the values used in Equations (1–3). [Color figure can be viewed at wileyonlinelibrary.com]

of Henshaw et al. (2020). For most LHDs, the upstream river width generally matched the dam width (W). However, for a few LHDs—where substantial variations in upstream river width were observed, a length-weighted average of river widths was used in Equation (1).

The height of sediment deposition (Z) was calculated as:

$$Z = H - D \tag{3}$$

where H is the height of the dam (m) and D is the average depth from the water surface to the sediment surface in the surveyed depositional zone (m), which extended up to 100 m upstream from each LHD.

The sediment depth ratio (S_{dr}), which represents the ratio of the dam's height filled by sediment, was calculated as:

$$S_{dr} = Z / H \tag{4}$$

The S_{dr} metric can be useful for comparing sediment deposition between LHDs of contrasting size and landscape position.

Upon determination of the stored sediment volume, the mass of sediment (kg) was calculated:

$$M_{LHD} = \rho \cdot V_{LHD} \tag{5}$$

where ρ is the bulk density of the sediment (kg/m^3). Hereinafter, sediment mass estimates are converted to, and presented in, units of metric tons.

With sediment storage volumes calculated, the potential impact of LHD storage on downstream reservoir sedimentation was determined. The downstream reservoir sedimentation rate (R_{sed}) represents the rate at which the multi-purpose pool of a reservoir accumulates sediment (m^3/year) and was retrieved from estimates published in USACE (2024). The fractional contribution to annual reservoir sedimentation (f_{ARS}) by the sediment stored behind an LHD, if it were to fail or be removed, was represented using the ratio:

$$f_{ARS} = V_{LHD} / R_{sed} \tag{6}$$

3.5 | Estimating Sediment Storage in Remotely Sensed LHDs

The basin-scale analysis included LHDs at least 1 m in height ($n = 19$) as identified by the Low Head Dam Task Force (2024). Seven were field-surveyed (Section 3.2), and an eighth (Marysville Dam) was included based on field measurements (Mansfield et al. 2024). The remaining 11 LHDs are hereinafter referred to as "remotely sensed" sites. For these, the same calculations were applied, substituting field-measured values with estimates from remote sensing or with imputed values from surveyed sites in the same basin. Uncertainty in substituted values was assessed with Monte Carlo analysis.

River slope and dam width were calculated using Google Earth. Dam height was estimated using LiDAR data from OpenTopography 3DEP products (United States Geological Survey 2021). As the water surface prevents LiDAR from penetrating to the streambed, remotely sensed LHD height was approximated as the difference between the upstream and downstream water surfaces. To assess the accuracy of the estimated height of remotely sensed LHDs, this geospatial method was applied to the surveyed sites, yielding good agreement ($R^2 = 0.91$, Figure 3); although LiDAR generally underestimates the true dam height.

Because the sediment depth ratio (S_{dr}) and bulk density (ρ) are unknown at remotely sensed sites, they were imputed from the range observed at surveyed LHDs within the same reservoir basin. In the Monte Carlo analysis ($n = 100{,}000$ realizations per site), LiDAR-estimated dam height was varied from 100% to 130% of the measured value (to compensate for underestimation; Figure 3), S_{dr} from 0.30 to 0.60 (based on field observations), and ρ between the minimum and maximum measured within the basin. For reach realization, V_{LHD} and M_{LHD} were computed, and results are reported as the mean plus-or-minus the standard deviation ($\mu \pm 1\sigma$) of all realizations.

To develop predictive models of V_{LHD}, power-law equations were fit one-at-a-time using variables that can be remotely sensed at LHDs—dam height, dam width, and river slope—as well as watershed-scale drivers, such as mean upstream precipitation (P_{mean}) and the watershed area ratio (A_{LHD}/A_{RES}). The terms A_{LHD} and A_{RES} represent the drainage areas of the LHD and reservoir, respectively. P_{mean} and A_{LHD}/A_{RES} were derived from the BasinATLAS dataset (Linke et al. 2020). Each power law regression, $V_{LHD} = a \cdot X^b$, was linearized using a logarithmic transformation, $\log(V_{LHD}) = \log(a) + b\log(X)$, where X, a, and b represent a predictor variable, a constant coefficient, and power coefficient, respectively. Statistical significance ($\alpha = 0.05$) and 95% confidence intervals were determined for the coefficients and regressions. All sites (surveyed and remotely sensed) were included in model fitting. Goodness of fit was assessed with the coefficient of determination (R^2).

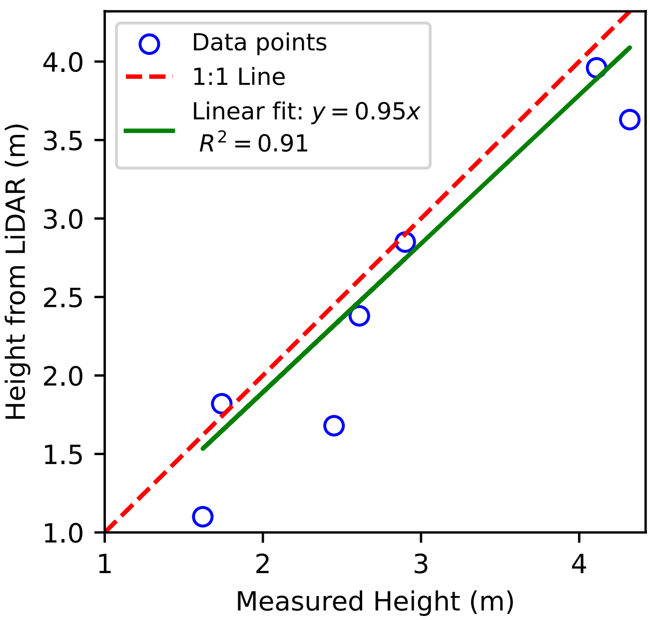

FIGURE 3 | Dam height comparison showing the strong agreement between remotely sensed LiDAR estimates and field survey measurements ($n = 7$). A line of best fit is shown, as is a 1:1 line. R^2 = coefficient of determination. [Color figure can be viewed at wileyonlinelibrary.com]

4 | Results

4.1 | Results for Surveyed Low-Head Dams

4.1.1 | Physical Characteristics of Low-Head Dams

A total of 7 LHDs were surveyed: one upstream from Perry Lake, three upstream from Tuttle Creek Lake, and three upstream from Kanopolis Lake (Table 1). River slopes at surveyed LHDs ranged from 0.04% (Holmesville, NE) to 0.11% (Hays, KS), increasing westward, consistent with regional topographic gradients. Dam heights ranged from 1.62 m at Russell, KS, to 4.32 m at Ellis, KS. Similarly, the dam widths ranged from 19.2 m at Hays, KS, to 48.9 m at Ellis, KS. The depth of sediment accumulation upstream of the LHDs ranged from 0.78 m at Kickapoo, KS, to 2.68 m at Ellis, KS. Sediment depth ratios (S_{dr}), representing the fraction of dam height filled with sediment, varied from 0.30 at Milford, NE, to 0.67 at Hays, KS. The Marysville Dam, which failed in 2018, was wider than the surveyed LHDs, but other physical properties fell within surveyed ranges.

4.1.2 | Sediment Grain Size and Bulk Density

Median grain size (d_{50}), bulk density, and porosity are summarized in Table 2. LHD deposits within the Tuttle Creek Lake basin were primarily silt, whereas Perry Lake and Kanopolis Lake basin deposits were fine sand (Figure 4). Mean d_{50} varied by basin, with LHD deposits in Tuttle Creek Lake being the finest (d_{50} from 24.5 to 27.7 μm) and those in Kanopolis Lake being the coarsest (d_{50} from 85.7 to 199.8 μm). Bulk density reflected grain size, with the lowest value at Holmesville, NE (874 kg/m^3) and the highest at Kickapoo, KS (1431 kg/m^3).

4.1.3 | Sediment Stored Behind Low-Head Dams

The volume (V_{LHD}) and mass (M_{LHD}) of stored sediment deposits varied as a function of physical dam and river measurements together with sediment properties (Table 2). Holmesville, NE, exhibited the highest sediment storage ($V_{LHD} = 360{,}900$ m^3), while Hays, KS, stored the smallest volume ($V_{LHD} = 8700$ m^3). The mass of sediment stored, calculated from the volume and bulk density, ranged from 315,400 tons (Holmesville, NE) to 10,400 tons (Russell, KS). Of the seven surveyed LHDs, the Holmesville, NE, site had the lowest river slope, largest dam width, and second-largest dam height, which together contribute to substantial sediment trapping and storage.

TABLE 1 | Field characteristics of surveyed low-head dams (LHDs) in Perry Lake, Tuttle Creek Lake, and Kanopolis Lake watersheds.

Downstream reservoir	LHD name	Latitude	Longitude	Slope (%)	Dam (m) Width	Dam (m) Height	Sediment depth (m)	Sediment depth ratio (S_{dr})
Perry Lake	Kickapoo, KS	39.660	−95.645	0.09	24.5	2.61	0.78	0.30
Tuttle Creek Lake	Washington, KS	39.808	−97.055	0.06	28.7	2.90	1.80	0.62
	Holmesville, NE	40.197	−96.663	0.04	48.9	4.11	2.42	0.59
	Milford, NE	40.740	−97.025	0.07	22.4	2.45	0.83	0.34
Kanopolis Lake	Ellis, KS	38.939	−99.555	0.10	43.7	4.32	2.68	0.62
	Hays, KS	38.873	−99.346	0.11	19.2	1.74	1.17	0.67
	Russell, KS	38.819	−98.960	0.08	20.5	1.62	0.96	0.59
Tuttle Creek Lake	*Marysville, KS*	*39.853*	*−96.664*	*0.08*	*61.5*	*3.70*	—	—

Note: Sediment depth represents the mean depth of sediment accumulation in the 100-m upstream of an LHD. In italics, the characteristics of a LHD that recently failed (see Mansfield et al. 2024) are shown as a comparison to the results of this study. Empty values (–) indicate that the variable could not be determined.

TABLE 2 | Measured and calculated sediment parameters of surveyed low-head dams (LHDs) in Perry Lake, Tuttle Creek Lake, and Kanopolis Lake watersheds.

Downstream reservoir	LHD name	d_{50} (μm)	Bulk density (kg/m^3)	Porosity (%)	Volume stored (×10^3 m^3)	Mass stored (×10^3 ton)	R_{sed} (×10^6 m^3/year)	f_{ARS}
Perry Lake	Kickapoo, KS	144.9	1431	47.6	8.7	12.5	1.61	0.005
Tuttle Creek Lake	Washington, KS	27.7	1058	50.8	82.8	87.6	3.33	0.025
	Holmesville, NE	28.2	874	64.2	360.9	315.4		0.108
	Milford, NE	24.5	884	66.7	12.0	10.6		0.004
Kanopolis Lake	Ellis, KS	199.8	1428	50.7	159.9	228.4	0.72	0.222
	Hays, KS	85.7	1419	40.3	10.2	14.5		0.014
	Russell, KS	86.8	902	62.1	11.6	10.4		0.016
Tuttle Creek Lake	*Marysville, KS*	—	—	—	*858.7*	—	*3.33*	*0.258*

Note: In italics, the parameters of a LHD that recently failed (Mansfield et al. 2024) are shown as a comparison to the results of this study. Empty values (–) indicate that the variable could not be determined. R_{sed} represents the annual multi-purpose pool sedimentation rate in the federal reservoir downstream of the LHD (USACE 2024). f_{ARS} represents the ratio of sediment stored behind an individual LHD to that of 1 year of sediment accumulation that occurs in the downstream reservoir.

The fraction of the annual downstream reservoir sedimentation (f_{ARS}) stored by individual LHDs ranged from 0.005 (Milford, NE) to 0.222 (Ellis, KS) (Table 2). The mean f_{ARS} for surveyed LHDs was 0.056, indicating that the surveyed LHDs store, on average, 5.6% of the annual downstream reservoir sedimentation amount.

4.1.4 | Depositional Bathymetry and Scour Holes

Bathymetric surveys revealed scour holes upstream of all LHDs (Figure 5), indicative of active sediment transport and reworking during high-flow events (Magilligan et al. 2021; Pearson and Pizzuto 2015; Rollet et al. 2024). Longitudinal centerline profiles show scour depths of 0.2–1.3 m relative to surrounding deposits. At one site (Kickapoo, KS), the scour holes extend beyond the impounded sediment and into the stream bed. The deepest point of the scour holes was typically positioned 20–50 m upstream of the dam face. Although scour of the deposits deviates from the assumed wedge geometry, their areal footprint (~50 m) is small relative to the backwater length (L on the order of ~1000 m). Consequently, wedge-based storage estimates are likely conservative upper bounds.

4.2 | Results for Remotely Sensed Low-Head Dams

4.2.1 | Storage Estimates for Remotely Sensed Sites

The eleven remotely sensed sites exhibit a range of river slopes, dam widths, and dam heights comparable to the surveyed sites (Table 3). Figure 6 shows the results of the Monte Carlo analysis for sediment volume and mass estimation for an LHD. Estimated V_{LHD} spanned 14,700 ± 3200 m^3 to 2,194,000 ± 471,900 m^3,

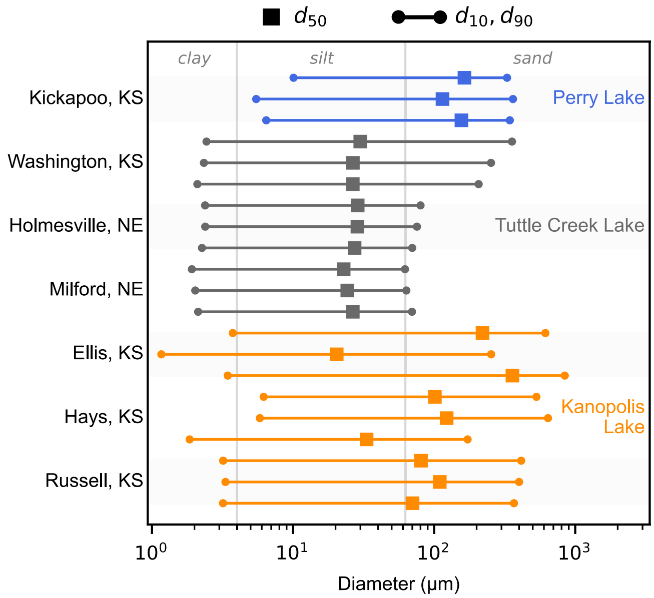

FIGURE 4 | Particle size distribution results for sediment cores collected upstream of LHDs, indicating the median (d_{50}), 10th percentile (d_{10}), and 90th percentile (d_{90}) diameters. Three distributions are shown for each site, and they represent subsamples of the same core. Sampling sites are colored based on the federal reservoir they drain into: Perry Lake (blue), Tuttle Creek Lake (gray), and Kanopolis Lake (orange). [Color figure can be viewed at wileyonlinelibrary.com]

corresponding to M_{LHD} values of $14{,}700 \pm 3200\,m^3$ to $2{,}194{,}000 \pm 471{,}900\,m^3$ (Table 4). Mean f_{ARS} of remotely sensed sites (0.112) slightly exceeded that for surveyed sites (0.056) (Figure 7).

4.2.2 | Predictive Equations for Sediment Storage

Power-law regressions relating V_{LHD} to H, W, and S (Figure 8) show that V_{LHD} increases with H and W and decreases with S. Coefficients for H and W were statistically significant ($p < 0.05$) while the coefficient for S was marginal ($p = 0.06$). The coefficients of determination (R^2) for the H, W, and S equations were 0.62, 0.58, and 0.19, respectively, indicating dam height and width are moderate predictors of sediment storage volume, while river slope is a marginal predictor. The basin-scale predictors, A_{LHD}/A_{RES} ($p = 0.29$, $R^2 = 0.07$) and P_{mean} ($p = 0.97$, $R^2 = 0.00$), were not statistically significant.

5 | Discussion

This study demonstrates that LHDs in major Kansas River basins impound considerable, albeit highly variable, volumes of sediment. The potential release of this stored sediment upon LHD failure or removal represents a relatively small, but nonetheless notable fraction of the annual sediment load delivered

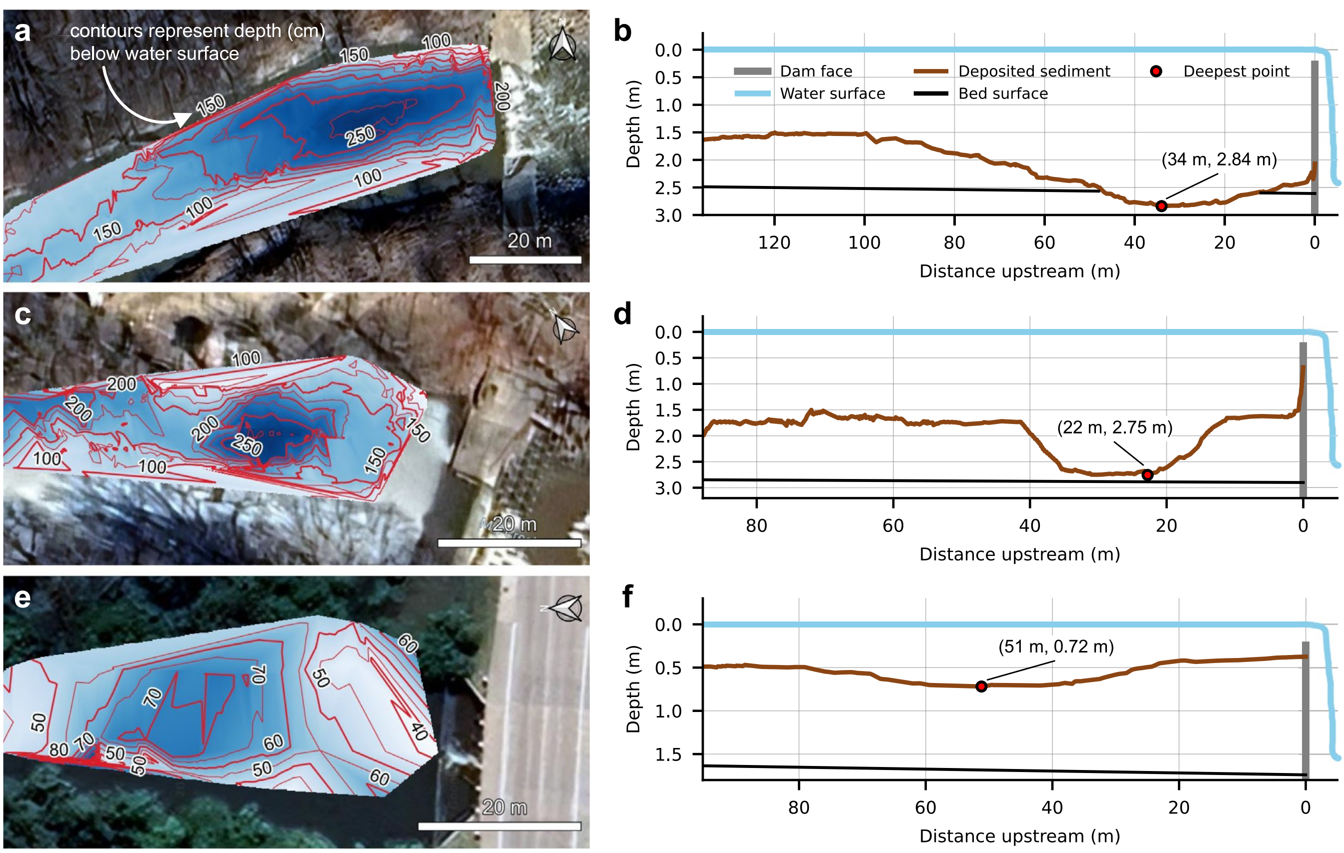

FIGURE 5 | Depth contour plots (a, c, e) and longitudinal centerline profiles (b, d, f) for the Kickapoo, Washington, and Hays low-head dams (LHDs), respectively. The deposited sediment surface was measured with the surveying vessel. The bed surface is inferred from mean channel slope and the depth to the river bed downstream of the LHD. Scour holes were observed upstream of each LHD, and their deepest point is noted. Flow direction is left to right in all panels. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 | Field characteristics of remotely sensed low-head dams (LHDs) in Perry Lake, Tuttle Creek Lake, and Kanopolis Lake watersheds.

Downstream reservoir	LHD number	Latitude	Longitude	Slope (%)	Dam (m)	
					Width	Height
Perry Lake	34	39.581	−95.554	0.09	48.5	1.61
Tuttle Creek Lake	28	40.131	−97.183	0.05	65.8	4.88
	15	40.053	−96.589	0.04	79.9	5.48
	46	39.490	−96.527	0.06	25.7	3.05
	21	40.090	−96.540	0.07	18.5	1.97
	19	40.098	−96.507	0.07	14.8	2.63
	20	40.096	−96.521	0.07	16.6	1.30
	26	40.078	−96.489	0.04	17.0	1.50
	12	40.266	−96.754	0.04	68.6	1.00
	32	40.989	−97.177	0.02	23.7	1.87
Kanopolis Lake	43	38.864	−99.334	0.11	22.7	2.44

Note: The LHD number is an identifier used in the Low Head Dam Task Force (2024) dataset.

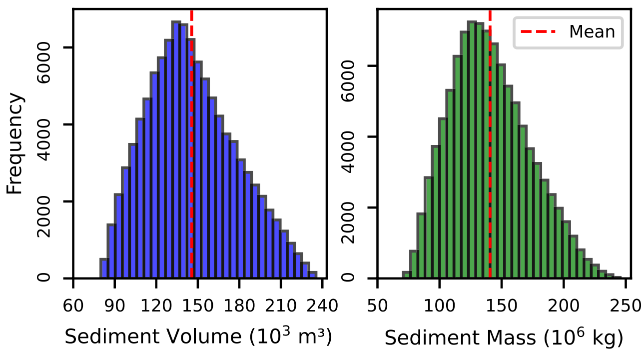

FIGURE 6 | A demonstration of the uncertainty results for estimating sediment (a) volume and (b) mass stored at a remotely sensed site (LHD #46) upstream of Tuttle Creek Lake. The mean of 100,000 Monte Carlo realizations is indicated by a vertical red line. [Color figure can be viewed at wileyonlinelibrary.com]

to downstream federal reservoirs. This potential was quantified using the f_{ARS} metric (fraction of annual reservoir sedimentation), which ranged from 0.005 to 0.66 for individual LHDs (Tables 2 and 4). The median f_{ARS} value was 0.025, indicating that a typical LHD stores the equivalent of 2.5% of the annual downstream reservoir sediment load. This variability highlights the need for site-specific assessments over generalized assumptions about LHD impacts.

5.1 | Basin-Scale Implications for Reservoir Sedimentation

The combined f_{ARS} from all LHDs upstream of Tuttle Creek Lake indicates that a theoretical instantaneous release would exceed the total annual sediment input to the reservoir's conservation pool ($\Sigma f_{ARS} = 1.28$). This is notable because Tuttle Creek Lake has already lost over half of its original storage capacity to sedimentation (Rahmani et al. 2018) and experienced accelerated infilling after the Marysville LHD failure ($f_{ARS} = 0.258$; Mansfield et al. 2024). Although stabilizing bank erosion has received considerable attention as a sediment management strategy in Tuttle Creek watershed (USACE 2023), the volume of sediment stored by LHDs (4.3×10^6 m³) exceeds the volume of sediment erosion that would be prevented by stabilizing all remaining 176 bank erosion hotspots in Kansas upstream of the reservoir (3.5×10^6 m³ over 30 years). These results highlight LHDs as potentially critical, yet often overlooked, components of watershed sediment budgets.

5.2 | Factors Influencing Sediment Storage and Predictability

Observed and estimated V_{LHD} was largely explained by dam and river geometry. The correlative modeling indicates that dam height (H) and width (W) were moderate predictors of V_{LHD} ($R^2 = 0.62$ and 0.58, respectively; Figure 8). Conversely, river slope (S) showed weak predictive power in this dataset ($R^2 = 0.19$), suggesting that within the range of slopes studied, dam geometry exerts a stronger control on trapped volume than bed gradient. Watershed-scale drivers (A_{LHD}/A_{RES} and P_{mean}) had little-to-no relation to V_{LHD}, with R^2 values of 0.07 and 0.00, respectively, despite considerable gradients in P_{mean} (548–903 mm/year) and A_{LHD}/A_{RES} (0.001–0.491). One explanation is equilibrium infilling: once a threshold storage is achieved, LHDs may maintain a dynamic balance and pass excess sediment downstream. Higher P_{mean} may potentially accelerate the time to equilibrium, but not the equilibrium storage volume, although measurements over time would be required to evaluate this hypothesis.

5.3 | Dynamic Equilibrium Storage and Scour Holes

Field surveys indicate partial infilling ($S_{dr} = 0.30$–0.67; Table 1), consistent with dynamic equilibrium wherein periodic

TABLE 4 | Measured and calculated sediment parameters of remotely sensed low-head dams (LHDs) in Perry Lake, Tuttle Creek Lake, and Kanopolis Lake.

Downstream reservoir	LHD number	Volume stored ($\times 10^3\,m^3$)	Mass stored ($\times 10^3$ ton)	R_{sed} ($\times 10^6\,m^3$/year)	f_{ARS}
Perry Lake	34	51 ± 11	73 ± 16	1.61	0.032
Tuttle Creek Lake	28	1146 ± 251	1107 ± 251	3.33	0.330
	15	2195 ± 479	2120 ± 478		0.659
	46	146 ± 32	141 ± 32		0.044
	21	38 ± 8	36 ± 8		0.011
	19	54 ± 12	52 ± 12		0.016
	20	15 ± 3	14 ± 3		0.004
	26	35 ± 8	34 ± 8		0.010
	12	63 ± 14	61 ± 14		0.019
	32	152 ± 33	146 ± 33		0.046
Kanopolis Lake	43	45 ± 10	52 ± 13	0.72	0.062

Note: The LHD number is an identifier used in the Low Head Dam Task Force (2024) dataset. Monte Carlo realizations of volume and mass stored are shown as the mean plus-or-minus one standard deviation ($\mu \pm 1\sigma$). R_{sed} represents the annual multi-purpose pool sedimentation rate in the federal reservoir downstream of the LHD (USACE 2024). f_{ARS} represents the ratio of sediment stored behind an individual LHD to that of 1 year of sediment accumulation that occurs in the downstream reservoir.

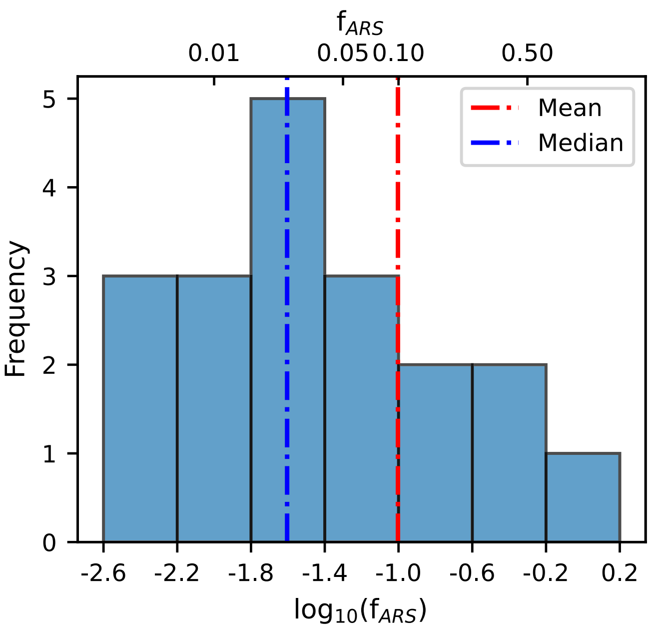

FIGURE 7 | Histogram of the fraction of annual downstream reservoir sedimentation (f_{ARS}) contained behind each LHD ($n = 19$). The mean (0.099) and median (0.025) f_{ARS} values are indicated by a vertical red and blue line, respectively. The sediment storage results include surveyed sites ($n = 7$), remotely sensed sites ($n = 11$), and the failed dam at Marysville, KS ($n = 1$). [Color figure can be viewed at wileyonlinelibrary.com]

flushing during high flows prevents complete infilling (Casserly et al. 2020; Ralston et al. 2021). The presence of scour holes upstream of the LHDs (Figure 5) provides further support for this concept as similar depositional features have elsewhere been linked to partial sediment continuity and episodic transport (Magilligan et al. 2021; Pearson and Pizzuto 2015). While a long-term net equilibrium implies limited year-to-year impact on downstream fluxes, the stored volumes represent latent potential loading should an LHD fail or be decommissioned. The presence of the scour holes reduces sediment depth in the vicinity of the LHD and causes divergence from the simple wedge geometry. However, localized scour zones occupy only a small fraction (~50 m) of the total backwater length (~1000 m), their influence on overall storage volumes is relatively small, and the assumed wedge geometry provides a conservatively high estimate for V_{LHD}.

5.4 | Staged Sediment Release and Management Opportunities

Proactive assessment of sediment volumes behind LHDs is essential for planning mitigation. The downstream impact of a dam failure or removal is typically not instantaneous. Instead, sediment is released in stages, a pattern observed at the failed Marysville Dam (Mansfield et al. 2024) and supported by conceptual models (Collins et al. 2017; Pearson et al. 2011). This process usually involves a rapid initial erosion phase followed by a slower, multi-year release. This staged dynamic creates a management opportunity. While the immediate impact to a downstream reservoir is lessened—for example, a 50% first-year release would halve the effect on Tuttle Creek Lake (Σf_{ARS} drops from 1.28 to 0.64)—a significant sediment pulse continues over several years. This prolonged release provides a crucial window to implement mitigation strategies, such as downstream sediment trapping or channel stabilization, to manage mobilized sediment before it permanently reduces reservoir capacity.

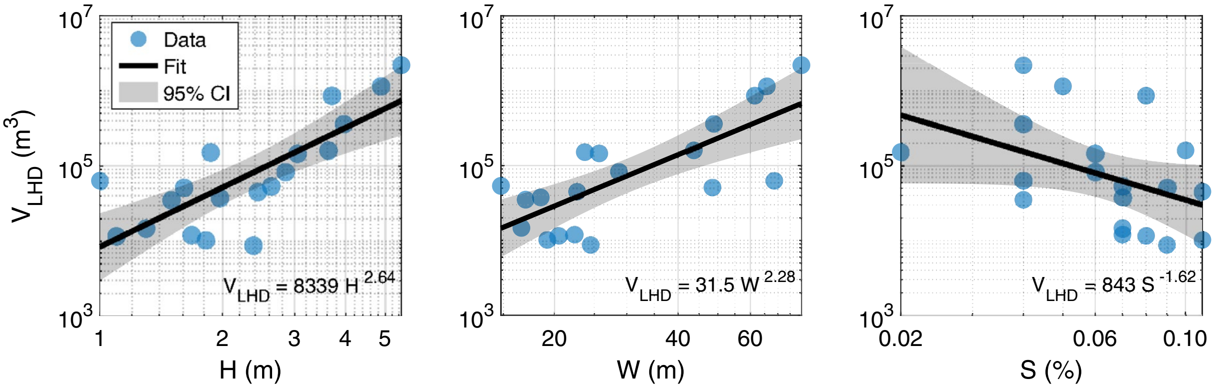

FIGURE 8 | Power law relations between the estimated sediment volume stored by a low-head dam (V_{LHD}) and the remotely sensed physical properties of LHDs: height (H), width (W), and river slope (S). The data points ($n = 19$) represent surveyed and remotely sensed low-head dams. The fitted equation, line, and confidence intervals (CI) are shown. [Color figure can be viewed at wileyonlinelibrary.com]

5.5 | Study Limitations and Future Work

Several limitations should be considered when interpreting these findings. Storage estimates rely on seven surveyed and eleven remotely sensed LHDs, and on simplified wedge-prism geometry. While necessary for basin-scale assessment, this approach introduces uncertainty related to the representativeness of the surveyed sites and the accuracy of remotely estimating key parameters like dam height (H), sediment depth ratio (S_{dr}), and bulk density (ρ). The geospatial dam height estimates, while correlated with field measurements ($R^2 = 0.91$, Figure 3), tended to underestimate true height, necessitating adjustment in the Monte Carlo analysis. Furthermore, imputing S_{dr} and bulk density values based on surveyed sites introduces uncertainty, as these properties can vary significantly.

Future research should aim to reduce these uncertainties. Expanding field surveys to a larger, more diverse set of LHDs across different physiographic settings would improve the robustness of storage estimates and predictive models. Refining remote sensing methodologies, potentially using higher-resolution topographic data (e.g., drone-based LiDAR) for dam geometry, and exploring geospatial proxies for sediment characteristics, could enhance estimation accuracy for unsurveyed sites. Longitudinal studies monitoring sediment transport following LHD removal or failure events are crucial for validating staged release models and quantifying downstream impacts more precisely. Integrating LHD storage estimates into comprehensive watershed sediment transport models would provide a more dynamic assessment of their role in sediment budgets and reservoir sedimentation under various flow and management scenarios.

6 | Conclusion

This study provides quantitative evidence that LHDs in Kansas contain significant, yet largely unquantified, volumes of stored sediment. The potential sediment release from these structures, whether through failure or planned removal, poses a quantifiable risk to the longevity of downstream federal reservoirs. This analysis shows that while a typical LHD stores a modest fraction of the annual downstream reservoir sediment load (median $f_{ARS} = 0.025$), the cumulative impact can be substantial. For example, in the case of Tuttle Creek Lake, the total sediment stored behind upstream LHDs is equivalent to more than one full year of reservoir sedimentation ($\Sigma f_{ARS} = 1.28$) and exceeds the estimated long-term sediment reduction from managing all upstream bank erosion hotspots in the State of Kansas.

These findings reveal that sediment storage volume (V_{LHD}) is strongly controlled by local dam geometry and river setting, not watershed-scale drivers like precipitation or drainage area. The influence of local factors suggests that many LHDs have reached a state of dynamic equilibrium with partial filling (30%–67% of dam height). The presence of scour holes just upstream of the LHDs, co-existing with large sediment storage volumes, indicates that these structures both promote latent storage and permit periodic sediment conveyance as flows accelerate over the dam.

From a management perspective, this latent storage is an important component to be considered in watershed sediment budgets. Even under a staged-release scenario, sediment mobilized by LHD failure or removal constitutes a substantial multi-year pulse to downstream systems. As such, incorporating LHDs into sediment management strategies is essential. While this assessment relies on modeling and a limited number of surveyed sites, it establishes a clear framework for future, more dynamic watershed-scale analyses. Ultimately, this work highlights the necessity of including LHDs in sediment management plans, providing the tools to prioritize high-risk structures and develop proactive strategies to enhance reservoir sustainability in Kansas and other river basins with aging infrastructure.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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